

ELECTRONIC CIRCUITS

MINI PROJECT (IC741)

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INTRODUCTION

This project demonstrates a simple temperature-activated circuit using an Op-Amp 741, a thermistor, and a trimpot.

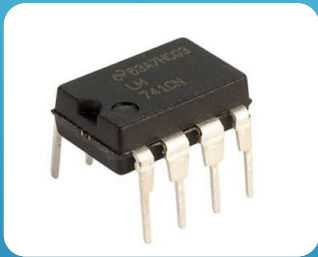
The goal:

The circuit is to turn ON a lamp when the temperature near the thermistor increases, such as when a flame or heat source is brought close.

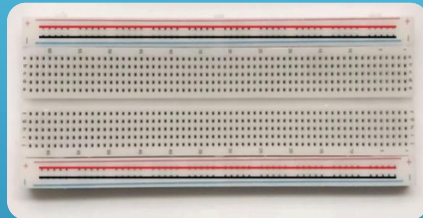
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Requirements:

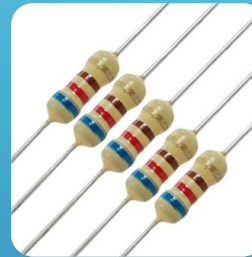
1- IC471



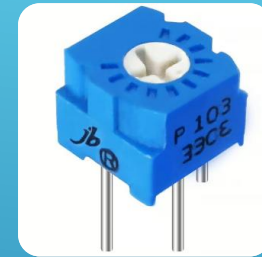
2- perbord



3- Resistors



4- potentiometer terminal



5- Thermal resistance



6- Wires



CIRCUIT DESCRIPTION

The circuit contains three important components:

1.Thermistor (NTC):

Its resistance decreases when temperature increases. This change in resistance creates a change in voltage.

2.Trimpot (Variable Resistor):

Used to set a reference voltage for the Op-Amp. This reference determines the temperature level at which the lamp will turn ON.

3.Op-Amp 741 (Comparator Mode):

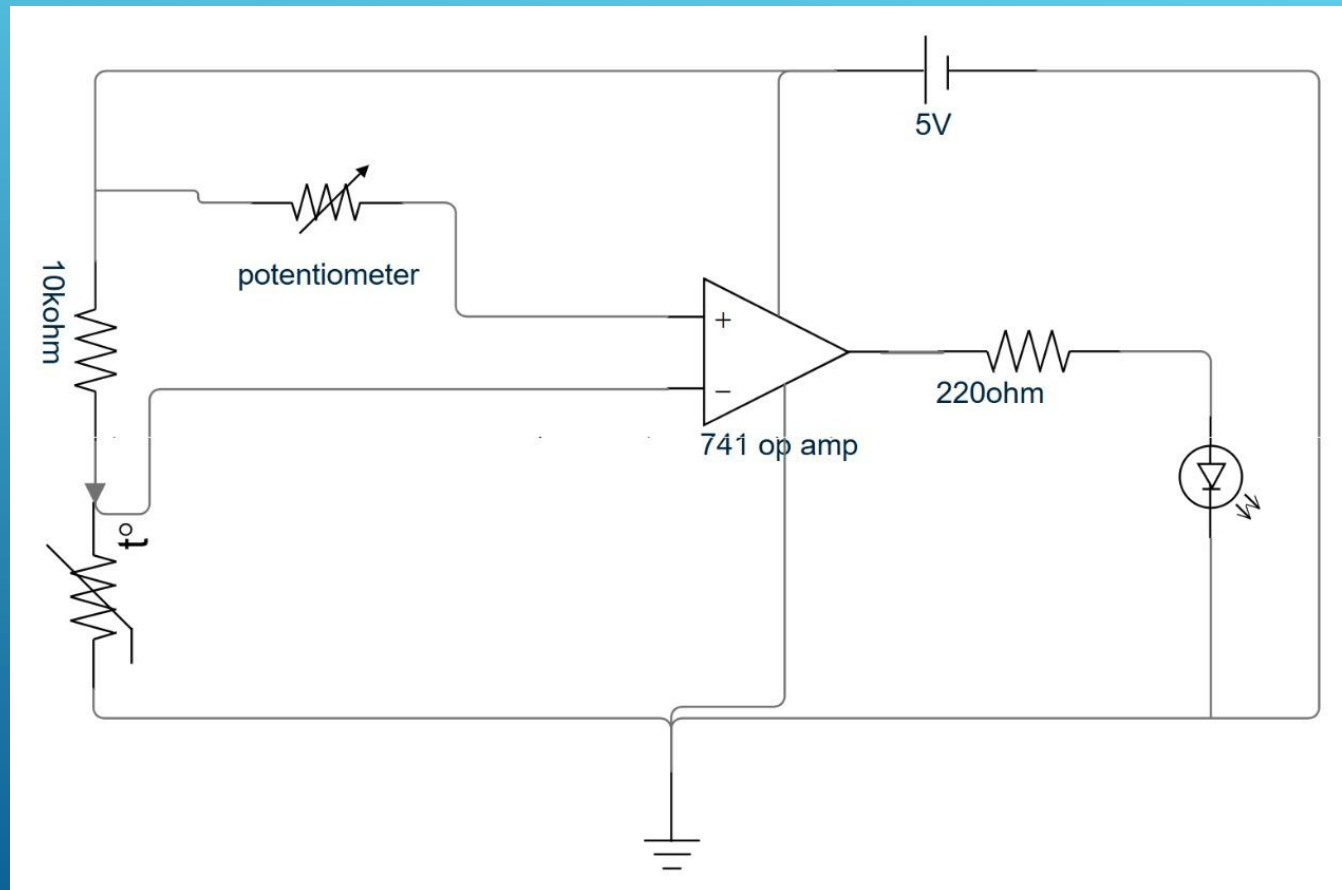
The 741 compares the thermistor voltage with the reference voltage set by the trimpot.

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WORKING PRINCIPLE

- The heat source (such as a lighter or candle) moves closer to the thermistor, the thermistor's resistance drops.
 - This produces a voltage that crosses the reference set by the trimpot.
 - At this point, the 741 output deactivates the lamp, indicating that the temperature threshold has been reached.
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- A decorative graphic consisting of several parallel white lines of varying lengths, arranged diagonally in the bottom right corner of the slide.

CIRCLE DRAWING



WHY DID WE USE IT IC741?

We used the IC 741 because it can detect small voltage changes from the thermistor, amplify them, and reliably control the lamp when the temperature rises.

Its simplicity and availability make it ideal for this experiment.

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USES

You can use this simple project as an early fire-detection indicator, where the LED turns off when heat is detected, helping alert you to potential fire hazards.

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IN THE FUTURE

If we developed it further, it could be used in real fire-alarm systems, smart home safety devices, or automatic emergency-response systems.

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CONCLUSION

This project is a simple and effective demonstration of temperature detection using basic electronic components.

The circuit works as a practical heat alarm or temperature-triggered switch.

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